

Anthropomorphism = the phenomenon of treating non-humans as if they are human, including attributing human forms to things or attributing human attitudes or abilities to things

Fake-Out Bot: A robot who exhibits misdirection by looking downward while, at the same time, using its camera to scrutinize a person that its gaze is avoiding (299)

Dishonest Anthropomorphism (299) =
A design that aims to exploit humans' deeply ingrained human reactions to human appearances and behaviors

What are the risks of dishonest anthropomorphism?

Frustrates ordinary processes of “boundary management” (301)

Boundary Management:

Humans manage their privacy by interpreting the environment around them, assessing the risk and opportunities it poses, and choosing strategies to deal with the possibility that others are observing and listening

e.g., through catfishing, phishing

Cybersecurity officers manage internet-based threats not by training users, but by imposing structural barriers that make it harder for users to be tricked (301-2)

e.g., biometrics, multi-factor authentication

Concedes that people are likely to be tricked

Why does dishonest anthropomorphism happen? (304)

- Sometimes to deceive people in order to abuse them
- Sometimes out of ignorance, including failure to understand how easily anthropomorphic tendencies arise

What Designers Can Abuse	Robot/AI Design Examples	Problematic Impacts of Design Choices
Human Responses to Body Parts	Installing cameras in a robot's neck or microphones in its chin, perhaps for increased functionality.	Human expectations about embodiment are subverted, which then distorts their assessment of the risks that a robot poses.
Human Responses to Body Language	Engineering a robot that cannot experience any emotions, to smile or frown, tilt its head to appear to commiserate, or jump back as if in surprise.	Humans intuitively infer that a robot is friendly, hostile, sympathetic, disinterested, etc., which distorts their ability to assess the risks that it poses or "nudges" them into attachment or trust.
Human Responses to Physical Behavior	Engineering a robot to pretend to sleep—closed eyes and curled up in the fetal position—while actively monitoring its surroundings. Engineering a robot to respect "personal space" boundaries to elicit more comfortable interactions.	Humans assume that a robot is conveying information about its behavior that would be true of a human and thus distorts their accurate assessment of the risks in their surroundings.
Human Responses to Living Beings	Engineering a robot to give the physical appearance of successfully eating and enjoying (or disliking) food that it can't digest or taste.	Humans become invested in a robot's wellbeing, including, perhaps, unhealthy levels of attachment or trust.
Human Responses to Size	Designing a robot to look slight and meek when it is physically powerful, perhaps to avoid intimidating users or just for reasons of space and costs.	Humans are unable to accurately judge what a robot can do physically, such as strength and speed, distorting their assessment of the risks that it poses.
Human Responses to Attractiveness (Appearance)	Engineering a robot to appear sexy, pleasant, wise, beautiful, handsome, and so on. (Or potentially, to appear unpleasant or ugly for some reason.)	Impacts how quickly humans are likely to try to please a robot and display other innate response to beauty; may nudge bonding, attachment and trust.
Human Responses to Voice	Engineering a robot that does not experience any emotion, to sound fearful or confident. Gendering the voice. Making it sound aged and wise, soothing and calm, or authoritative. Determining when to display excitement.	Humans assume a robot is conveying human-compatible information about non-existent attitudes and experiences. This distorts their assessment of the risks that it poses and may cause them to repose inappropriate levels of trust and reliance on its inputs.
Human Responses to Age	Designing a robot to look like a sweet and innocent child, a mature adult, or an enfeebled (or wisely aged) senior citizen.	Human judgment of what a robot can do cognitively or physically distorts their assessment of the risks that it poses; particularly if a childlike image engenders muttering by adults or untoward trust from children.
Human Responses to Personality (related to Identity, below)	Engineering a robot to express likes and dislikes, to appear engaging, authoritative, or submissive when it has no preferences or objective experiences.	Humans respond as they would to a likeminded or dissimilar human, and cannot prevent the manipulation of their reaction to what it communicates.
Human Responses to Agency	Engineering a Wizard of Oz trick to make a robot under partial or full human control seem autonomous.	Humans will inaccurately assess the threats a robot poses, over- or underestimate the robot's capabilities, and fail to appreciate the true context of their situation.
Human Responses to Identity	Engineering robots to appear gendered, to be of a certain race or culture, reflect social class, or particular skills.	Humans will engage with a robot by reflecting their own positive or negative prejudices; making them vulnerable to predatory purposes, or inciting inappropriate attachment or abusive behavior.
Human Responses to Localized Intelligence (related to Agency)	Engineering a smart and networked robot to appear as if it only processes and stores information locally or on-device.	Humans will over- or under-estimate the scope of robot capabilities, which will distort their assessment of the threat it poses.
Human Responses to Sensory Manipulation	Engineering a robot to control the environmental flow of sensory data (light, sound, possibly others) to influence a person's physical state (adjust light to control mood; sound to control focus or concentration)	Humans will be unknowingly vulnerable to sensory-based manipulation of their environment, impacting their mood, emotion, and behavior in response to unanticipated robot controls.
On Mobility	Engineering a programmed robot to look like it can move around on its own, while also inadvertently nudging the user to treat the robot as a living thing.	Humans will display increased anthropomorphic behaviors, potentially without countervailing benefit.

Figure 1: Taxonomy of Dishonest Anthropomorphism